ImQMD Raw Data Analysis Report

Deuteron Breakup Observables for DPOL Experiment

tbt

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Outline

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ImQMD Simulation Data

- ImQMD (Improved Quantum Molecular Dynamics) transport model
- Simulates deuteron breakup on various nuclear targets at $T_{\text{lab}} = 190$ MeV/u
- Outputs: proton & neutron momenta $(p_x^p, p_y^p, p_z^p, p_x^n, p_y^n, p_z^n)$, impact parameter b, reaction plane angle ϕ_{RP}
- Two polarization configurations:
  - **ypol**: tensor polarization along y-axis (perpendicular to beam)
  - **zpol**: tensor polarization along z-axis (beam direction)
- 10^6 events per configuration, symmetry energy parameter $\gamma \in [0.5, 0.85]$

Dataset Summary

Dataset	Targets	γ values	Polarizations
ypol (old)	^{208}Pb, ^{130}Xe	0.50, 0.60, 0.70, 0.80	ynp, ypn
ypol (20260413)	^{112}Sn, ^{124}Sn	0.50–0.85 (step 0.05)	ynp, ypn
zpol	^{208}Pb, ^{112}Sn, ^{124}Sn, ^{140}Xe	0.50–0.80	znp, zpn

Data format (`dbreak.dat`):

- ypol: 9 columns — event#, p_x^p, p_y^p, p_z^p, p_x^n, p_y^n, p_z^n, b, ϕ_{RP}
- zpol: 7 columns per impact parameter file — event#, p_x^p, p_y^p, p_z^p, p_x^n, p_y^n, p_z^n

Data Processing Flow

1. **Read**: parse `dbreak.dat` $\rightarrow$ Event objects $(p_x^p, p_y^p, p_z^p, p_x^n, p_y^n, p_z^n, b, \phi_{\mathrm{RP}})$
2. **Derive**: compute total momentum $\vec{P}_{\mathrm{tot}} = \vec{p}^p + \vec{p}^n$, azimuthal angle $\phi = \mathrm{atan2}(P_y^{\mathrm{tot}}, P_x^{\mathrm{tot}})$
3. **Rotate**: transform into reaction plane frame by rotating by $-\phi$ around z-axis

$$\begin{pmatrix} p_x' \\ p_y' \end{pmatrix} = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} p_x \\ p_y \end{pmatrix}$$

4. **Cut**: apply kinematic selections (loose $\rightarrow$ mid $\rightarrow$ tight)
5. **Accumulate**: collect momenta distributions and compute observables

Event Selection Cuts

ypol cuts (polarization axis $= y$):

Cut	Conditions	Motivation
loose	$\lvert p_y^p - p_y^n \rvert < 150,\ P_{xy}^2 > 2500$	unphysical breakup removal
mid	loose $+ (p_x'^p + p_x'^n) < 200$	in-plane momentum constraint
tight	mid $+ (\pi - \lvert\phi\rvert) < 0.2$	near-backscatter selection

zpol cuts (polarization axis $= z$):

Cut	Conditions	Motivation
loose	$(p_z^p + p_z^n) > 1150,\ \lvert p_z^p - p_z^n \rvert < 150$	forward + unphysical removal
mid	loose $+ (p_x^p + p_x^n) < 200,\ P_{xy}^2 > 2500$	transverse constraint
tight	mid $+ (\pi - \lvert\phi\rvert) < 0.5$	near-backscatter selection

Deuteron binding energy: $E_b = 2.2$ MeV $\Rightarrow$ max physical relative momentum ~ 45 MeV/c.
Cut at 150 MeV/c conservatively removes unphysical ImQMD breakups.

The R Observable

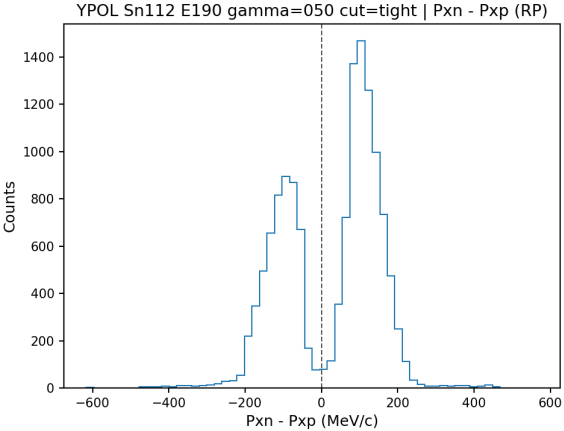
Definition

$$R = \frac{N(P_x'^p > P_x'^n)}{N(P_x'^p < P_x'^n)}$$

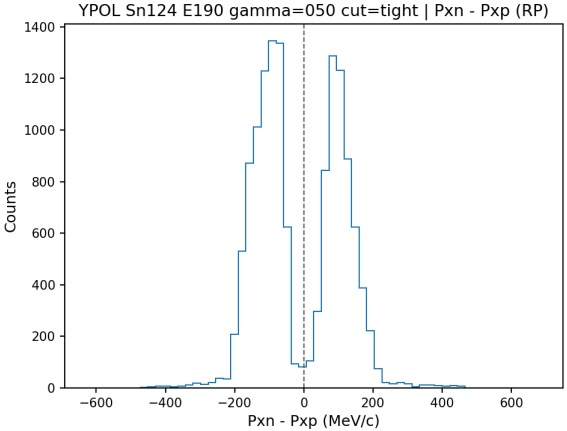
where $P_x'^{p,n}$ are proton/neutron x-momenta in the reaction plane frame.

- Quantifies the **isovector reorientation effect**
- In a non-uniform nuclear mean field, protons and neutrons experience different impulses
- R shows strong sensitivity to the symmetry energy parameter γ
- Key observable for constraining the nuclear equation of state

$P_x^p - P_x^n$ Distribution Example



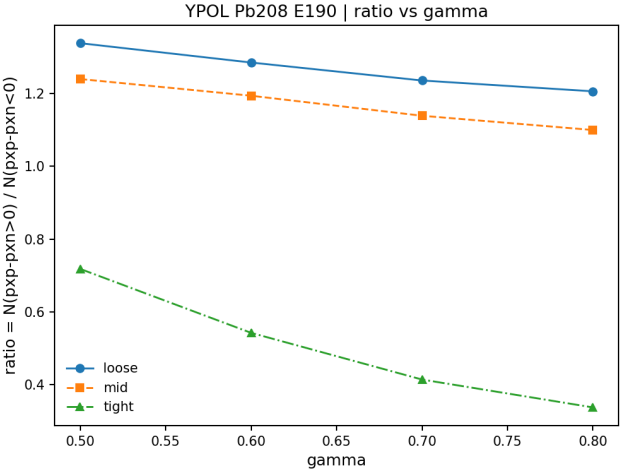
^{112}Sn, $\gamma = 0.50$, tight cut



^{124}Sn, $\gamma = 0.50$, tight cut

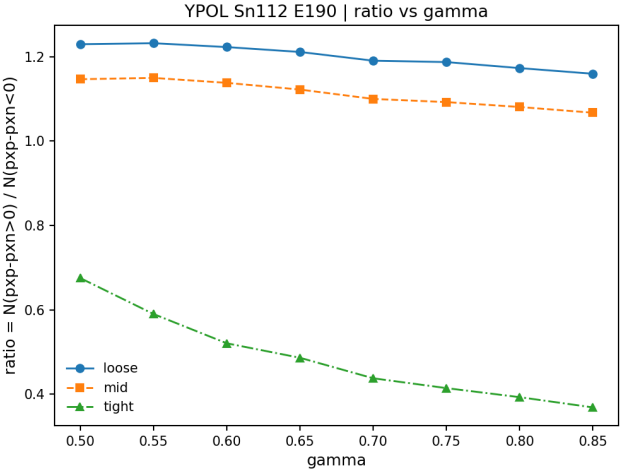
Asymmetry in the $P_x^n - P_x^p$ distribution directly reflects the isovector reorientation effect.

ypol R vs γ — ^{208}Pb



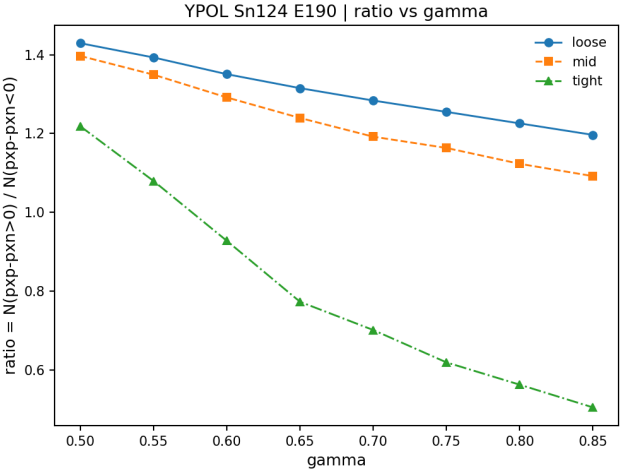
- Target: ^{208}Pb (old dataset)
- $\gamma = 0.50, 0.60, 0.70, 0.80$
- Tight cut: R drops from 0.72 to 0.34
- Strong γ sensitivity confirmed
- Tight cut enhances asymmetry signal

ypol R vs γ — ^{112}Sn (New Data)



- Target: ^{112}Sn (20260413 dataset)
- $\gamma = 0.50$–0.85 (8 points)
- Tight cut: R from 0.68 to 0.37
- $\sim$250k events/gamma (loose)
- Smooth monotonic decrease
- Finer γ grid resolves curvature

ypol R vs γ — ^{124}Sn (New Data)



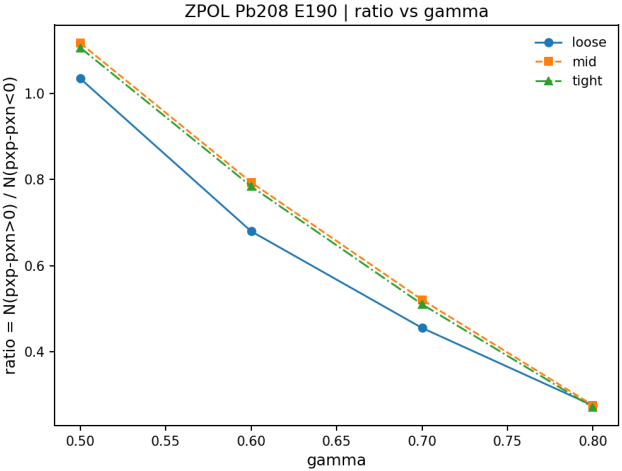
- Target: ^{124}Sn (20260413 dataset)
- $\gamma = 0.50$–0.85 (8 points)
- Tight cut: R from 1.22 to 0.50
- **Strongest γ sensitivity**
- Larger N/Z ratio amplifies isovector effect
- R crosses unity at $\gamma \approx 0.55$

ypol Comparison: R at Tight Cut across Targets

γ	^{208}**Pb**	^{112}**Sn**	^{124}**Sn**
0.50	0.718	0.675	1.218
0.55	—	0.590	1.080
0.60	0.542	0.521	0.928
0.65	—	0.486	0.773
0.70	0.414	0.438	0.701
0.75	—	0.414	0.619
0.80	0.338	0.393	0.562
0.85	—	0.369	0.504

- ^{124}Sn has the widest R range: from 1.22 to 0.50 ($\Delta R = 0.71$)
- ^{208}Pb: $\Delta R = 0.38$; ^{112}Sn: $\Delta R = 0.31$
- Neutron-rich ^{124}Sn is the most sensitive probe of γ

zpol R vs γ — ^{208}Pb



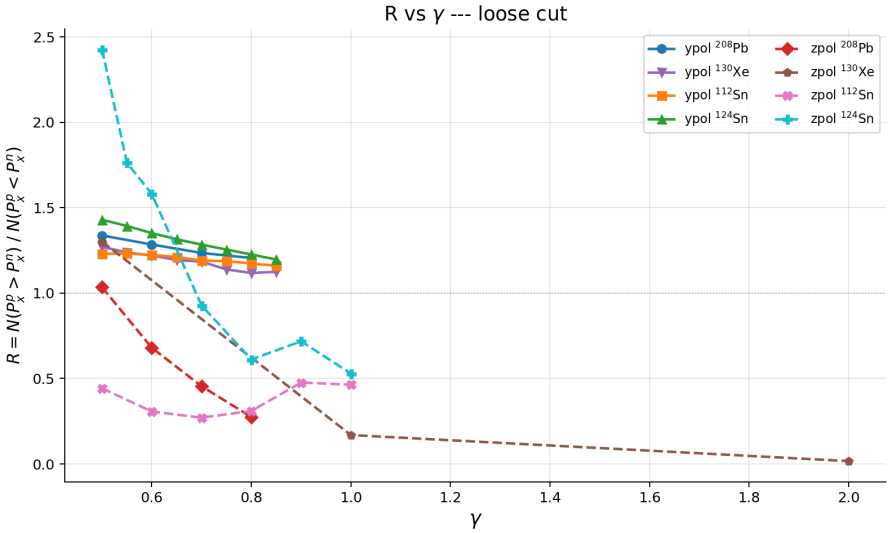
- Target: ^{208}Pb, zpol dataset
- $\gamma = 0.50, 0.60, 0.70, 0.80$
- Tight cut: R from 1.11 to 0.27
- **Very strong sensitivity**
- R drops by factor $\sim 4\times$
- zpol shows larger dynamic range than ypol for Pb

ypol vs zpol: ^{208}Pb Tight Cut

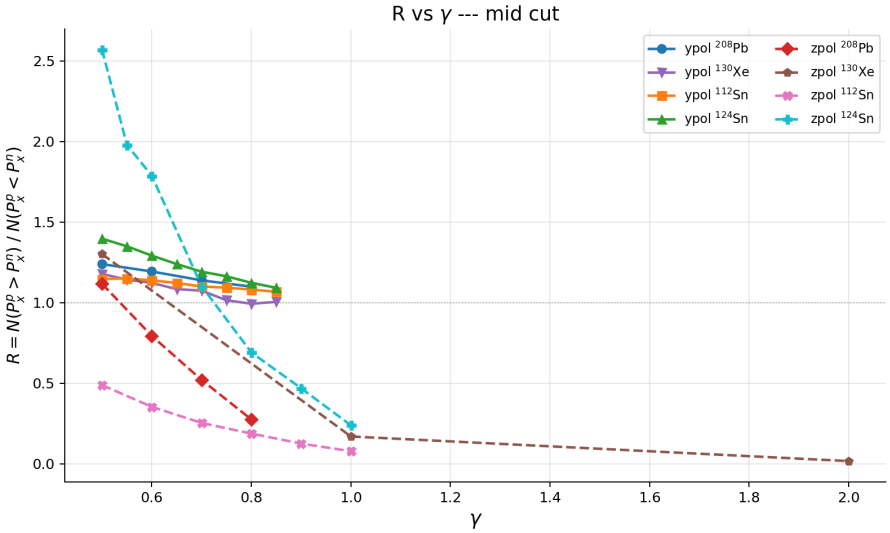
γ	**ypol** R	**zpol** R
0.50	0.718	1.106
0.60	0.542	0.784
0.70	0.414	0.510
0.80	0.338	0.270

- zpol: R ranges from 1.11 to 0.27 ($\Delta R = 0.84$) — crosses unity
- ypol: R ranges from 0.72 to 0.34 ($\Delta R = 0.38$) — stays below 1
- Different polarization axes probe complementary aspects of the symmetry energy
- Combined measurement constrains γ more tightly

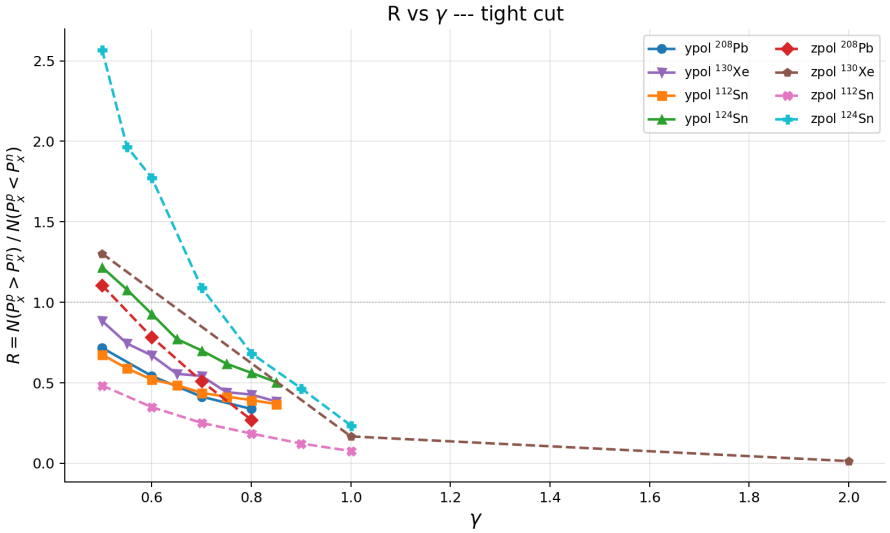
R vs γ — All Targets, Loose Cut



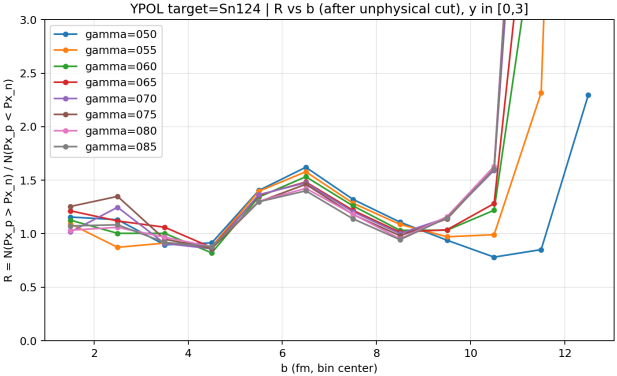
R vs γ — All Targets, Mid Cut



R vs γ — All Targets, Tight Cut

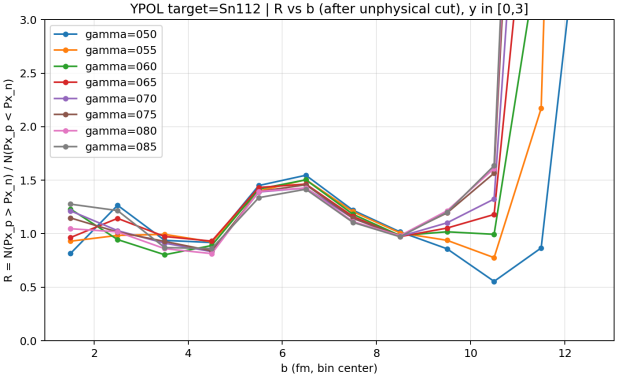


R vs Impact Parameter b — ypol ^{124}Sn



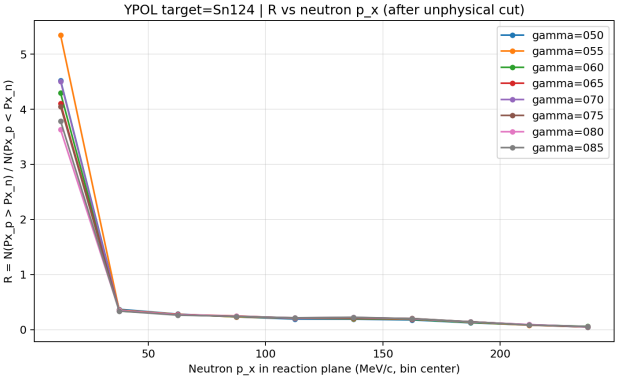
- $R(b)$ for different γ values
- γ separation strongest at large $b > 10$ fm (peripheral collisions)
- Central collisions ($b < 5$ fm): $R \approx 1$ for all γ
- Physics: peripheral collisions have longer interaction time, amplifying symmetry energy effect

R vs Impact Parameter b — ypol ^{112}Sn



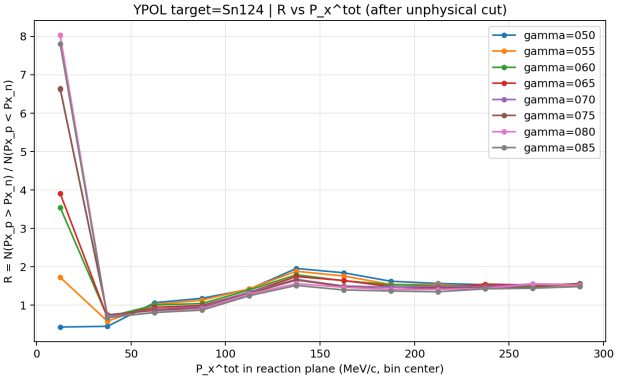
- Same trend as ^{124}Sn
- Smaller γ spread (lower N/Z)
- Peripheral region ($b > 10$ fm) still shows best separation
- Consistent with isovector mechanism

R vs Neutron p_x — ypol ^{124}Sn



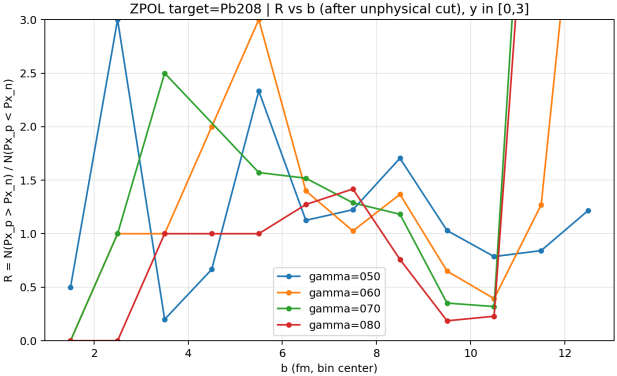
- R as a function of neutron p_x in reaction plane
- Strong γ dependence at low $|p_x^n|$
- High $|p_x^n|$ region: $R \to 0$ (neutron dominates)
- Sensitivity concentrated at $|p_x^n| < 50$ MeV/c

R vs Total P_x — ypol ^{124}Sn



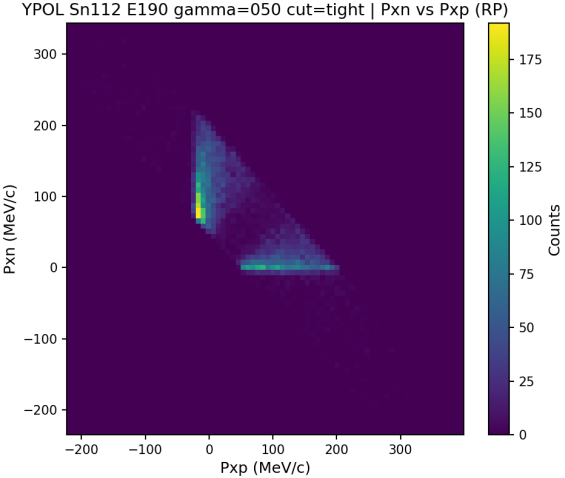
- R vs total in-plane momentum $P_x^{\text{tot}} = p_x'^p + p_x'^n$
- Best γ separation at low P_x^{tot}
- Consistent with tight cut physics: near-backscatter events with small total transverse momentum

R vs b — zpol ^{208}Pb

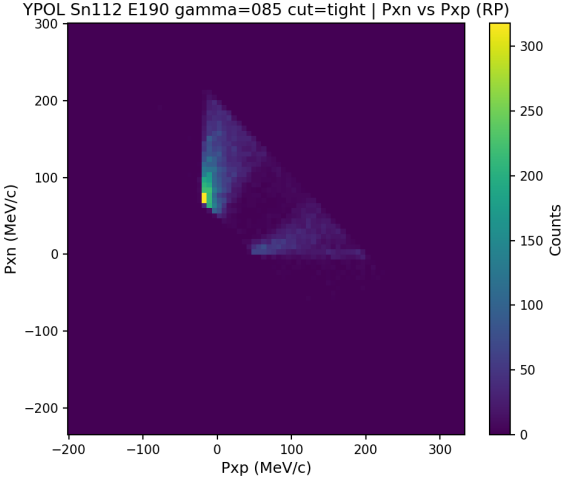


- zpol ^{208}Pb: $R(b)$
- Dramatic γ separation at $b > 8$ fm
- zpol probes beam-axis polarization response
- Complementary to ypol differential behavior

P_x^n vs P_x^p Correlation — ypol ^{112}Sn



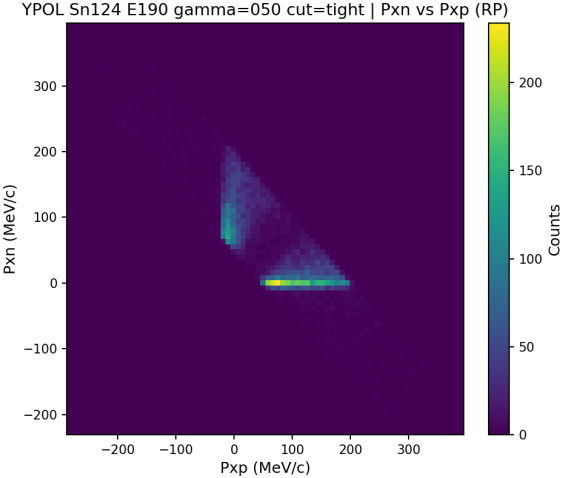
$\gamma = 0.50$



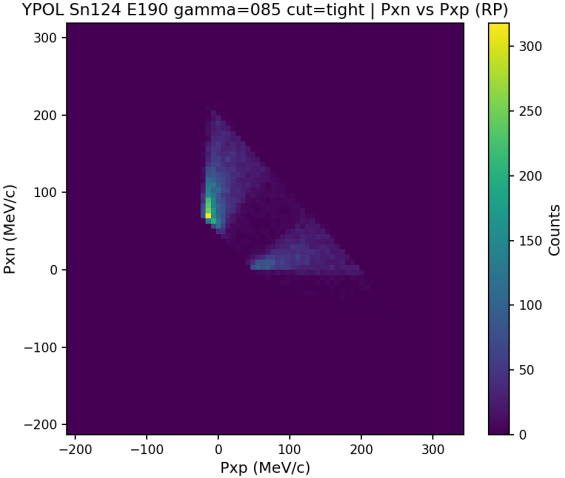
$\gamma = 0.85$

Reaction-plane rotated P_x^n vs P_x^p (tight cut). The correlation pattern shifts with γ, reflecting

P_x^n vs P_x^p Correlation — ypol ^{124}Sn



$\gamma = 0.50$



$\gamma = 0.85$

^{124}Sn shows more pronounced shift in the correlation pattern compared to ^{112}Sn, consistent

Summary

- Analyzed ImQMD deuteron breakup data for multiple targets and polarization configurations
- The $R = N(P_x^p > P_x^n)/N(P_x^p < P_x^n)$ observable shows strong sensitivity to γ

Dataset	**Target**	R **range (tight)**	ΔR
ypol	^{208}Pb	$0.72 \to 0.34$	0.38
ypol	^{112}Sn	$0.68 \to 0.37$	0.31
ypol	^{124}Sn	$1.22 \to 0.50$	**0.71**
zpol	^{208}Pb	$1.11 \to 0.27$	**0.84**

- Neutron-rich ^{124}Sn provides the strongest ypol sensitivity
- zpol ^{208}Pb has the largest overall dynamic range
- Differential $R(b)$: peripheral collisions ($b > 10$ fm) dominate γ separation
- New 20260413 data (8 γ points) resolves curvature in $R(\gamma)$